

Deep Q-Learning on Atari Pong

Hyperparameter Experiments Summary | Emmanuella Briggs

Drive Link : https://drive.google.com/drive/folders/1LZhpmMPnav2wToNz26UbVt8kACGpQRQJ?usp=drive_link

Experiment	Policy	Learning Rate	Gamma	Batch Size	Epsilon Decay	Mean Reward (Overall)	Noted Behavior
Emma_exp 01	CNN	2.5e-4	0.99	32	0.1	-20.82	Baseline run, still losing almost every point at this stage
Emma_exp 02	MLP	5e-4	0.99	256	0.1	-20.94	Almost no improvement, flattening pixels loses spatial information
Emma_exp 03	CNN	5e-4	0.99	256	0.1	-18.93	Best performing run, clearly outperforms the matching MLP run in exp02
Emma_exp 04	CNN	1e-4	0.999	32	0.1	-20.86	Higher gamma alone gives no improvement over baseline
Emma_exp 05	CNN	1e-4	0.99	128	0.1	-20.63	Larger batch size gives a mild improvement
Emma_exp 06	CNN	1e-4	0.99	256	0.1	-19.86	Larger batch size continues to help, second best overall
Emma_exp 07	CNN	5e-5	0.99	256	0.1	-20.41	Lower learning rate slows learning, slightly worse than exp06
Emma_exp 08	CNN	1e-4	0.99	32	0.05	-20.79	Faster epsilon decay gives no meaningful change
Emma_exp 09	CNN	1e-4	0.99	32	0.3	-20.76	Slower epsilon decay gives no meaningful change
Emma_exp 10	CNN	1e-4	0.999	256	0.1	-20.35	Combining higher gamma and larger batch underperforms batch size alone